SCOPE OF WORK
50’x100’x18’ Red Iron Storage Facility
MCAAP, Oklahoma 74501-9002
27 August 2026

1.0 PROJECT OVERVIEW

The Contractor shall provide all supervision, labor, materials, and equipment to perform the tasks defined in this scope of work and any included specifications, drawings, or other contract documents for the installation of the Storage facility. Contractor must coordinate all final measurements, material lists, construction documents and quantities needed with the Government prior to initiation of work.

All work shall be in accordance with reference drawings, Scope of Work (SOW), all applicable Codes and regulations, and any other data provided for technical requirements. The following description provided is general in nature and not all-inclusive in indicating the work necessary to complete the scope of work. It is the responsibility of the contractor to plan and provide the elements necessary for completion. Contractor will be on site along with government and any required sub-contractors to make final inspection and acceptance.

This Scope of Work (SOW) outlines the requirements for the construction of a 50’x100’x18’ Red Iron Storage Facility and its associated site improvements, utilities, electrical distribution, communications, and security infrastructure. The contractor is responsible for executing all works in compliance with the work categories outlined below.

- **Utility Clarification:** Please note that MCAAP will execute all disconnects, reconnects, and terminations of all utilities. The contractor is responsible for coordination and preparation of pathways, trenching, and infrastructure up to the designated tie-in and termination points.

1.1 Pre-Construction, Permitting & Safety

1.1.1 Engineering & Site Analysis

- **Geotechnical Analysis:** The contractor shall perform a comprehensive geotechnical soil analysis at the construction site. This analysis must cover the building footprint, the proposed perimeter fence lines, and all planned road areas.

1.1.2 Permits & Regulatory Compliance

- **Excavation Permits:** Secure all necessary Dig Permits and coordinate utility locates prior to any ground-breaking, trenching, or excavation activities. *(Applies to site preparation, electrical, grounding, and fencing activities).*

Job Hazard Analysis (JHA):

- All personnel must read, understand, and sign the Job Hazard Analysis (JHA) prior to starting any on-site work.
- The signed JHA must be maintained at the job site for the duration of the project and returned to the supervisor immediately upon job completion.

1.2 Site Preparation & Civil Works (Earthworks)

1.2.1 Clearing & Demolition

- **Site Clearing:** Clear the proposed 3.5-acre construction site of all trees, brush, and vegetation.

- **Fencing Perimeter Clearing:** Clear a 12-foot zone around the outside perimeter of the proposed new fencing, and a 30-foot zone inside the fence line.

1.2.2 Grading & Site Utilities Trenching

- **Bulk Grading & Excavation:** Grade and compact the entire construction site area. Excavate precise footprints for all concrete slabs and footings, including:

  - Main building slab and footings ($50' \times 100' \times 8''$)
  - Walkthrough door stoop ($4' \times 4' \times 4''$)
  - Front slab apron ($8' \times 100' \times 8''$)
  - Eastern loading pad ($45' \times 40' \times 8''$)

- **Trenching Coordination:**

  - Excavate trenches required for the Lightning Protection System (LPS) and the building grounding loop.
  - Excavate the utility trench running from the transformer pole to the storage building.

1.2.3 Access Road Construction

- **Gravel Loop Road:** Construct a $575' \times 25'$ gravel road connecting to the main asphalt road in a continuous loop.

  - The road must route directly in front of the storage facility and continue past the second proposed storage facility site within the loop.
  - **Subgrade Stabilization:** Lay a structural geogrid between the subgrade and subbase course if soil conditions dictate or if required by the geotechnical analysis.

1.3 Concrete Works, Foundation, & Subgrade Grounding

1.3.1 Subgrade Grounding & Welding (Pre-Pour)

- **Rebar Structural Bonding:** Weld necessary rebar joints together for the $50' \times 100' \times 8''$ and $8' \times 100' \times 8''$ concrete slabs as required for grounding and bonding purposes.

- **Building Grounding Loop:**
  - Install a continuous grounding loop around the building footprint for structural and Lightning Protection System (LPS) grounding.
  - Use **1/0 19-strand soft-drawn bare copper wire** buried at a minimum depth of 30 inches below grade.

• **Exothermic Welding:** Cad-weld four (4) joints of the 1/0 ground wire directly to the building foundation's corner rebar.

- **Corner Ground Rods:**

- Install two (2) copper ground rods at each of the four corners of the building (8 ground rods total), buried 12 inches below grade and spaced 6 feet apart.
- Position rod sets between 3 feet and 8 feet away from the building exterior.
- Install a $24'' \times 24'' \times 12''$ precast concrete ring ground rod box over each ground rod for inspection access.

1.3.2 Concrete Placement & Finishing

• **Main Monolithic Slab & Footings:** Prep, form, and place ten (10) $4' \times 4' \times 3'$ grade-on-beam concrete column footings, integrated into a $50' \times 100' \times 8''$ monolithic concrete slab.

• **Mix & Reinforcement:** Use 4,000 PSI concrete reinforced with #5 (5/8") rebar spaced 12 inches On Center, Each Way (O.C.E.W.).
• **Finish:** Finish slab surface with a smooth steel trowel finish.
• **Control Joints:** Saw-cut control joints in the main slab to a depth of 1/4 of the slab thickness (2 inches deep) forming $12.5' \times 12.5'$ square sections O.C.E.W.
• **Concrete Stoop (West Entrance):** Prep, form, and place a $4' \times 4' \times 4''$ concrete stoop on the west side of the building. This stoop must sit directly in front of the $3' \times 7'$ walkthrough door, attach directly to the main building slab, and receive a non-slip broom finish.
• **Concrete Apron (Front):** Prep, form, and place an $8' \times 100' \times 8''$ monolithic concrete apron running the entire length of the building's front (South) side. The apron must attach directly to the main building slab and receive a broom finish.
• **Concrete Loading Pad (East):** Prep, form, and place a $45' \times 40' \times 8''$ monolithic concrete loading pad with integrated footings on the East side of the building (refer to drawings for exact alignment).
• **Apron Subbase Transition:** Backfill and compact base course gravel into the transition zone between the concrete apron and the new gravel road.

1.3.3 Bollard Installation

• **Safety Bollards:** Install sixteen (16) $7' \times 6''$ safety yellow concrete-filled steel bollards.

- **Placement:** Position in pairs flanking each of the four (4) overhead doors (both interior and exterior locations).
- **Embedment:** Encase the bottom 3 feet of each bollard directly within the concrete slab/footing pours.

1.4 Structural Steel & Exterior Building Enclosure

1.4.1 Red Iron Building Structural Erection

- **Primary Frame:** Erect a $50' \times 100' \times 18'$ Pre-Engineered Red Iron Steel Building with a 4:12 roof pitch.

- **Wall & Roof Panels:**

  - Install **26-gauge R-panel** wall sheets.
  - Install **24-gauge R-panel** roof sheets.
- Install all associated exterior color-matched trims, flashings, and closures.

- **Insulation:** Insulate the entirety of the building's roof and walls using **R-19 closed-cell spray foam insulation**.

1.4.2 Doors & Openings

- **Overhead Doors:** Install four (4) $22' \times 14'$ insulated steel, manual chain-driven overhead doors along the front (South) elevation.
- **Walkthrough Door:** Install one (1) $3' \times 7'$ heavy-duty steel walkthrough door equipped with a security hasp on the West elevation.

1.4.3 Daylighting & Drainage Accessories

- **Translucent Panels:** Install twelve (12) $3' \times 6'$ translucent poly-ribbed wall panels for natural interior lighting:

  - Two (2) panels on the West end wall, positioned just below the ridge line.
  - Two (2) panels on the East end wall, positioned just below the ridge line.
- Eight (8) panels on the North (back) wall, positioned just below the eave line.

- **Gutter System:** Install $5'' \times 5'' \times 18'$ galvanized steel box gutters along both the front (South) and rear (North) eave lines. Provide two (2) downspouts on the front and two (2) downspouts on the rear.

1.5 Electrical Utilities & Power Infrastructure

1.5.1 Overhead Power & Pole Line Distribution

- **Utility Pole Line Construction:**

  - Install nine (9) 35-foot, Class 3 steel utility poles equipped with 8-foot crossarms and pin insulators.
  - String **#6 solid hard-drawn bare copper wire** across the poles.

- **Alignment:** Route seven (7) poles parallel to the asphalt road (Stanton Road) spaced 170 feet apart. Route the remaining two (2) poles to the West, spaced 100 feet apart.
- **Structural Support:** Install down-guy wires and anchors on strain/dead-end poles as structurally required.

- **Transformer Installation:**

- Mount one (1) 25KVA 12470/7200V (Primary) to 120/240V (Secondary), single-phase, oil-filled overhead transformer (complete with voltage taps and fuse cutout) on the 35-foot pole closest to the building.
- Ensure the transformer pole is properly grounded to code.

1.5.2 Lightning Protection System (LPS) Poles

- **LPS Poles:** Install two (2) 55-foot, Class 3 steel (slip-joint) Lightning Protection System (LPS) poles complete with air terminals.

  - Position one pole centered on the exterior of the East end wall, and the other centered on the West end wall.
  - Connect bare copper grounding conductor from the poles to the underground loop system.

1.5.3 Service Entrance & Panel Distribution

- **Main Service Panel (Pole):**

  - Install a new 225A, 120/240V, 1-phase, NEMA 3R rainproof panel with a 100A main circuit breaker on the new 35-foot transformer utility pole.
  - Wire the service entrance using 2/0 THHN copper conductors routed through 2-inch rigid metal conduit (RMC) with a weatherhead.

- **Pole Grounding:** Install two (2) $8' \times 5/8''$ copper-clad ground rods, buried at least 6 feet apart, to ground the 225A main panel.

- **Subpanel Distribution (Buildings 1PM11 & 1PM12):**

  - Bury 2-inch rigid conduit from the pole panel to the exterior ends of both storage buildings (1PM11 and 1PM12).
  - Install a 100A, 120/240V, 1-phase, NEMA 3R subpanel on the exterior end of both buildings.
  - Pull 1/0 copper conductors to feed the subpanels' 50A main breakers (oversized to mitigate voltage drop over distance).

1.5.4 Interior & Exterior Building Power / Lighting

- **Exterior Lighting:** Run a dedicated lighting circuit to feed four (4) 40-watt exterior LED wall pack fixtures equipped with integrated photocells. Mount these fixtures on the South wall, centered in between each overhead door.
- **Conduit & Controls:** Run 3/4-inch EMT conduit to the Remote Area Data Collector (RADC) security box to supply dedicated utility power.
- **Door Switches:** Install magnetic door switches on the main walkthrough door and all four (4) overhead doors.
- **Branch Circuits:** Ensure all building light fixtures and Intrusion Detection System (IDS) panels are run on dedicated 20A branch breakers.

1.6 Communications & Fiber Optic Infrastructure

- **Aerial Fiber Optic Run:** Route 24-strand, single-mode, indoor/outdoor-rated aerial fiber optic cable from across the asphalt road. Attach it to the first new electrical utility pole, run it to the second pole (100-foot span), and transition underground.
- **Underground Fiber Pathway:** Route the 24-strand fiber optic cable inside underground PVC conduit from the utility pole splice box directly into the building footprint for IDS use.
- **Communications Termination:** Route and terminate the fiber cable inside the building interior adjacent to the $3' \times 7'$ walkthrough door. House the terminations within a $48'' \times 48'' \times 12''$ metal enclosure containing the primary network switch and the security RADC cabinet.

1.7 Security Systems & Intrusion Detection (IDS)

- **Intrusion Detection System (IDS) Wiring:**
    - Wire new RADC panel box and all magnetic intrusion contacts.
    - Install one (1) heavy-duty magnetic contact on the walkthrough door and four (4) floor-mounted armored magnetic contacts on the overhead roll-up doors.

- **Wiring & Conduit:** Pull **18-2 single-pair shielded red riser cable** from the IDS panel to all magnetic contacts. Ensure all security wiring is fully encased in protective steel EMT conduit.

1.8 Perimeter Security & Fencing

1.8.1 Fencing Installation

- **Main Perimeter Fence:** Install 1,000 linear feet of 6-foot-high chain-link perimeter security fencing and commercial gates as shown on drawings.
- **Barbed Wire topping:** Attach industrial 45-degree outriggers to the tops of all fence posts and install a 3-strand barbed wire security topping.
- **Fencing Alignment:** Run the fence line continuously along the West side of the buildings, extending from the NW corner behind building to the SW corner behind the proposed building location, squaring off with existing perimeter corner posts.

1.8.2 Security Gap Seating (Dirt work)

- **Gap Seal Requirement:** Provide all necessary grading, backfilling, and localized dirt work below the fence line to install concrete lintels or rebar gap sealing. This detail must be executed to strictly satisfy the 2-inch maximum ground-to-fence security gap requirement.

1.9 Signage, Site Cleanup & Project Closeout

1.9.1 Building Identification

- **Signage:** Fabricate and install a custom building identification sign on the exterior facade, centered directly above the two middle overhead doors.

- *Note: The contractor must coordinate directly with the PW Sign Shop to verify and obtain exact sign dimensions and letter specifications prior to fabrication.*

1.9.2 Daily Maintenance & Demobilization

- **Daily Cleanliness:** Clean and organize the active work area at the end of each shift or working day to maintain a safe, professional workspace.
- **Final Clean down:** Perform a comprehensive cleanup of the entire job site upon completion of all structural, civil, electrical, and security works. Remove all debris, excess materials, and temporary safety barriers.
- **JHA Handover:** Sign off on the completed JHA paperwork and return it directly to the Project Supervisor to initiate final project inspection and acceptance.

2.0 LOCATION

McAlester Army Ammunition Plant is located 6 miles south of McAlester, OK in Pittsburg County, Oklahoma. Storage site is in a secure area within the installation.

3.0 BASIS FOR PAYMENT

Progress payments will be made only upon completed portions of the project and materials on site.

4.0 WORK HOURS

Required work hours will be from 0600 – 1630 Monday through Thursday, excluding federal holidays. Requests for adjustments to working hours and/or days of operation must be clearly defined and requested with a minimum of one week's notice for the contracting officer or designated representative Any changes must be approved through appropriate channels.

5.0 COMPLETION TIME

All work, to include cleanup, final inspection and acceptance by the Government, and submission of all post-construction documentation is 180 working days from issuance of notice to proceed. This will account for any lead time estimated for shipping of purchased equipment, as well as installation.

6.0 SECURITY & STAFF REQUIREMENTS

All contractors and subcontractors must be U.S. Citizens AND have a Real ID or compliant document to access the installation. This includes but is not limited to bids, site visits, execution of projects, installation and prove outs of equipment, necessary environmental testing, delivery and removal of material or equipment, etc. Security from MF190-20 is required to be submitted no less than three working days prior to access for full background information to be verified.

7.0 SAFETY REQUIREMENTS

The contractor shall comply with the current Corps of Engineers (USACE) Safety Health Requirements Manual, EM 385-1-1 and all applicable federal, State, and local safety codes and practices. It is the contractor's responsibility to ensure public safety during the entire duration of the contract. Necessary signs, barriers, warning devices, etc. shall be operated and maintained by the contractor at contractor expense. Contractor employees shall comply with the minimum dress requirements as stated in Chapter 5 of EM 385-1-1. The contractor shall submit a safety hazard analysis for approval before commencing work. Work near live MCAAP operations may require

additional safety measures. MCAAP safety will provide guidance as necessary.

8.0 EQUIPMENT

All equipment used to perform work under this contract shall be in good operating mechanical condition and shall comply with manual EM 385-1-1. Equipment shall be inspected by the contractor, and inspection reports shall be submitted and approved by the contracting officer or contracting officer representative (COR) prior to utilization of such equipment to accomplish work on this contract.

9.0 SUBMITTALS

The contractor shall be required to furnish certain submittals. These submittals shall be certifications of materials, safety plans, and work plans. Also, specific materials found in any data provided. These will be specifically listed and shall be furnished to the contracting officer representative before or at the pre-work conference. The work plan shall identify a detailed execution schedule that least impacts the availability of the facility to the end users. The safety plan shall identify precautions that will be implemented keeping the park users free from any types of safety hazards. Copies of approved work plan and safety plan will be on site during the duration of this task order. Submittals requiring government approval shall be formally prepared and approval granted prior to proceeding with work involving these submittals. FIO (for information only) submittals do not require government approval. The contractor shall submit manufacture's recommended installation instructions prior to work.

10.0 PRE-WORK CONFERENCE

Prior to the commencement of any work, a pre-work conference will be held with the project manager, contracting officer or designated representative, contractor, and anyone else the Contracting Officer deems necessary. The contractor's person responsible and necessary administrative staff will be required to attend this meeting. Proper procedures, invoicing process, payment, schedule, material storage, etc. will be covered at this time. A detailed phasing plan will be necessary from the contracting project manager for review at this time.

11.0 QUALITY ASSURANCE

It shall be the responsibility of the contractor to ensure quality control and assurance measures are defined and in place for material and workmanship and reporting is provided to the Government as specified. It shall be the responsibility of the contractor to provide any equipment, work, and reporting necessary for QA review by the JOC inspector. All work activities by the contractor requiring inspection and monitoring shall be coordinated and scheduled with the Government and inspector to ensure the inspector is present for those activities.

All processes must comply with the following regulations. When regulations are contradictory, the regulation that is more stringent is the one that is most applicable.

12.0 SITE CLEANUP

The contractor shall always maintain a clean work area. The contractor shall be responsible for the protection of the environment and will be required to restore all damages subject to the approval of the contracting officer. The contactor shall remove and dispose of all debris off government property in a state approved disposal site unless specified otherwise. The contractor will be required to comply with all Federal and State environmental requirements. The contractor shall remove all excess materials and clean the entire area. The cleanup shall be completed daily.

13.0 POINT OF CONTACT

The point of contact will be the Contracting Officers Representative for ASMG-PWP-(JOC) Sarah Howard, 918-420-6861, sarah.e.howard37.civ@army.mil

14.0 PRODUCTS

The contractor shall furnish all materials and supplies required to complete the work.

15.0 MATERIALS

All materials will be in accordance with MCAAP provided specifications. All materials must be submitted on MCAAP form 200-23-CONTRACTOR'S DECLARATION OF MATERIALS, CHEMICALS, OR SUBSTANCES at least one week prior to execution for environmental review and documentation. Materials must meet environmental requirements and standards and not be used until an official approval has been received.

15.1.DELIVERY AND STORAGE

Materials shall be delivered to the site undamaged and stored out of contact with the ground. Materials that are damaged during delivery or improper storage shall be replaced at no cost to the government. The Contracting Officer shall inspect all materials once on site, prior to commencement of work.

16.0 EXECUTION

16.1. ACCOMPLISHMENT OF WORK

The Contractor shall provide all equipment and material to construct the designed project. All work shall comply with the project MCAAP specifications. The contractor shall inform the Contracting Officer's Representative (COR) of any conflicting or incorrect portions prior to causing a delay to allow the GOV to provide an acceptable solution. All work shall comply with established codes and regulations.